A 3.1.1 Activity

Activity: Relational vs. Non-Relational Scenarios

Objective:

To distinguish between scenarios best suited for relational databases and those better suited for non-relational databases.

Instructions:

In your groups, you will be given a set of scenario cards. Each card describes a situation where a database is needed.
Your task is to analyse each scenario and decide whether a relational or non-relational database would be more appropriate.
For each scenario, provide a justification for your choice. Consider the specific requirements of the scenario, such as the type of data, the need for relationships between data, scalability requirements, and the importance of data consistency.
Be prepared to present your decisions and justifications to the class.

Investigation Structure:

To guide your investigation, consider the following structure for each scenario:

1. Scenario Summary: Briefly describe the main purpose of the database in the given scenario.
2. Data Characteristics:
What type of data needs to be stored (e.g., structured, unstructured)?
Are there complex relationships between different pieces of data?
How important is data consistency and integrity?
What are the scalability requirements (e.g., will the database need to handle large amounts of data or many users)?
3. Database Choice:
Would a relational or non-relational database be more suitable?
4. Justification:
Explain why you chose the database type in step 3.
Discuss the advantages of your chosen database type for this scenario.
Address any potential drawbacks or limitations.

Scenario 1:

Description: A social media platform needs to store user profiles, posts, comments, and connections between users.
Considerations: Relationships between users, unstructured data (posts, comments), scalability.

Scenario 2:

Description: An e-commerce website needs to store customer information, product details, orders, and inventory.
Considerations: Transactions, data consistency, complex queries for reports.

Scenario 3:

Description: A scientific research project collects large amounts of sensor data, including timestamps and measurements.
Considerations: Time-series data, scalability, simple queries.

Scenario 4:

Description: A content management system needs to store articles, images, and user comments.
Considerations: Flexible schema, unstructured data, ease of updates.

Scenario 5:

Description: A bank needs to store customer accounts, transactions, and financial data.
Considerations: Data integrity, security, complex transactions.